What goes in, and what do we predict?

Current position: the pipeline trains and its predictions respond to news and attention. A reliable forecasting advantage from news has **not yet been established**.

1 / Data and target

Public CMIN-US release: headline titles linked to companies, source dates and daily prices/volume. Our fixed cohort contains **12 companies** and **9,984 eligible company-days** from 2018 to April 2021.

Each forecast uses **five trading sessions**, with up to six retained headlines per arrival bucket. Frozen sentiment encoding supplies positive, neutral and negative scores.

Target: at today's close, predict whether the *next trading session's adjusted close* will be higher. UP if higher; DOWN otherwise.

Source news date D is assumed available from $D+2$ at midnight UTC. Exact historical publication/availability is uncertain. Headlines are selected by a fixed hash rule, not by relevance.

A real example / Apple (AAPL)

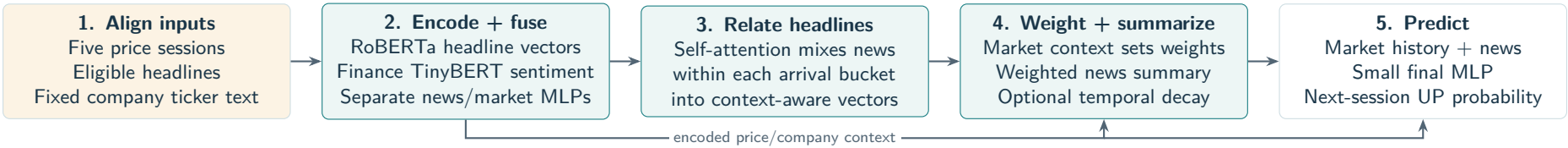
Stored headline: "Apple Earnings Preview: How Will the Coronavirus Impact iPhone Sales?"

Source date: **28 April 2020**. Proxy arrival: **30 April**. Forecast after the 30 April close; target: **1 May**.

Observation	Adjusted close
30 April 2020	\$72.5467
1 May 2020	\$71.3787

Actual movement: -1.61% **/ DOWN**. Saved baseline model: **47.38% UP probability / DOWN**. This example was correct; **one example is not performance evidence**. Future prices enter the target only.

2 / Simplified pipeline



Frozen: RoBERTa and sentiment models. **Learned:** text projection, price/sentiment encoders, fusion layers, attention and final MLP. Prices supply the market query and also enter the predictor directly.

2a / What actually enters the final MLP?

Version	Input to the final predictor	Size / history
FININ paper, Eq. 7	Encoded index history (prices + market description) + daily weighted news summaries (Reuters headlines + instrument sentiment)	1–20 input days; width depends on model
Our initial baseline	Five market vectors + five daily news summaries	$5 \times 16 + 5 \times 16 =$ **160**
Our recent temporal tests	Five market vectors + one news vector pooled across the five buckets	$5 \times 16 + 16 =$ **96**

Inside these vectors: market = prices/volume + ticker context; news = headlines + sentiment, mixed and weighted. Raw sentiment, headline weights and ages are **not separate final-MLP columns**; they shape the news summary upstream.

Does news reach the prediction? What do interactions do?

3 / Direct evidence of news and weight dependence

Hold the **trained network and market inputs fixed**; change only the named news input or attention operation. No retraining in these probes. The examples below use the **160-input baseline**, forecast date **31 August 2020**, seed 17.

Company	Original UP probability	Controlled change	New UP probability
AEP	47.73% (DOWN)	Replace headline vectors with their training mean; retain prices, sentiment and masks	54.49% (UP)
AEP	47.73% (DOWN)	Bypass headline self-attention; retain existing trained weights	55.56% (UP)
AAPL	53.64% (UP)	Replace market-query headline weights with uniform weights	56.69% (UP)

Across four selected networks and **44 news-containing validation model-cases**, headline replacement changed 44/44 probabilities; uniform query weighting changed 16/44. These are illustrative sensitivity checks, not retrained forecasting comparisons. The original outputs reproduced saved predictions.

What this establishes: the model uses news and attention. **What it does not establish:** meaningful understanding of the news, or better forecasts. Input substitutions can move a prediction in the wrong direction.

5 / How headline interactions work

Each headline starts as a vector containing text and sentiment. Self-attention replaces it with a learned mixture of the headlines in its bucket. Market-query weighting then combines those **mixtures**. Interactions run separately within each bucket, not directly between individual headlines across days.

Inspecting the Apple example on page 1:

Title topic (shortened)	Final bucket weight
Dow Jones earnings; Apple buy point	16.5%
Expert warns of stock-market selling	13.6%
Autos and the chip industry	18.1%
Dow Jones and consumer confidence	15.9%
SAP/Deutsche Telekom contact-tracing app	20.2%
Apple earnings and iPhone sales	15.6%

These six retained titles mix direct Apple events with broad/other-company topics. This is a title-only inspection; article bodies and discarded stories were not assessed. The largest weight is on a *contextual mixture*, so it is not proof that the SAP story itself matters most.

New inspection / Does mixing preserve distinctions?

We inspected the four saved full-model traces. The table measures how similar headlines' vectors are **before and after self-attention**: a value near 1 means almost the same direction.

Period / seed	Before	After
2020 / 17	0.418	0.9998
2020 / 42	0.530	0.9997
2021 / 17	0.216	0.3665
2021 / 42	0.834	1.0000

Finding: in three of four inspected networks, mixing makes headline vectors almost identical in direction. This raises a **blending/bottleneck concern**, but does not prove useful information was erased. Vector lengths can still differ; the fourth model behaves differently.

Median within-bucket pairwise cosine similarity; 34 multi-headline buckets per model, from two saved first-evaluation snapshot dates. The reconstructed attention outputs match saved vectors. This is a diagnostic, not a semantic or causal test.

New comparison (four fits): remove *only self-attention*; retain market-query weighting and the same inputs/training settings. Full loss: **0.690388**; no-interaction loss: **0.693387**. Removal increases pooled loss by **+0.002999**, conditional 95% interval **[+0.001058, +0.005092]**. The pooled comparison favors interactions, but period means disagree and the average effect is below the 0.005 practical criterion. Full still trails prices (0.690095).

News decay: a negligible effect, with unresolved causes

4 / Completed temporal tests

Matched inputs: same eligible news, five price sessions and 96-input pooled MLP. Equal weighting is compared with a fixed **48-hour half-life**: every extra 48 hours halves a story's multiplier before normalization. This is news-age weighting, not optimizer weight decay.

Evaluation: 2,004 company-days on 167 distinct dates; May–August 2020 and January–April 2021; two seeds. Lower log loss means better probabilities.

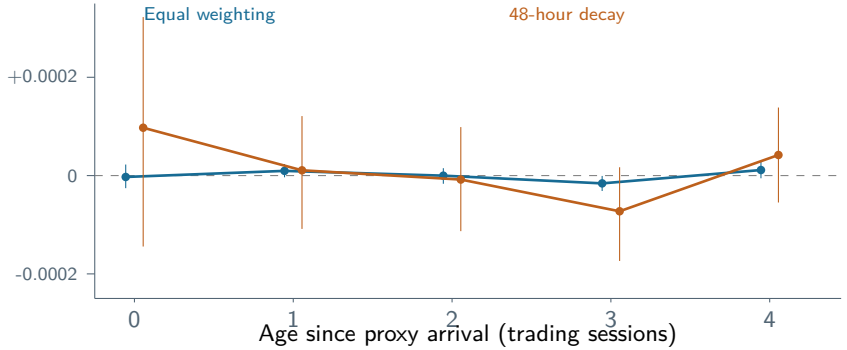
Model / reference	Log loss
Training-prevalence probabilities	0.689820
Prices only (matched pooled geometry)	0.689771
Prices + current arrival bucket	0.689651
Prices + current and older news, equal weighting	0.690210
Prices + current and older news, 48-hour decay	0.689887

Decay improves loss by only **0.000324**. Its conditional 95% difference interval is **[**-0.001176**,** $+0.000539$**]**, crossing zero. The predeclared practical improvement of 0.005 was not reached. Adding older buckets also failed to improve forecasting reliably.

Interpretation: treat the decay change as **practically negligible and indistinguishable from noise**. Do not interpret tiny curve bumps as a market discovery.

What does the age curve measure?

Remove one age bucket at a time, keep prices/model fixed, and predict again. Plot **loss after removal minus original loss**.



Above zero: removal worsens loss. **Below zero:** removal improves loss. Age 0 is the current *arrival* bucket, not necessarily freshly published news. Bars are conditional 95% intervals.

Result: tiny, non-monotonic effects; **9 of 10 intervals include zero**. Removing news also renormalizes remaining weights. The curve does not estimate an economic news half-life.

Normalized decay tests relative recency: a common age offset cancels. FININ itself has no explicit exponential-decay term; this is an independent extension.

Why might useful forecasting information be missing?

Candidate	What we observed	What is still unproven
Timing / relevance	Date-based D+2 eligibility; mixed headline topics; six-story hash cap	Whether useful information is already priced in or useful stories are excluded
Aggregation / learning	Near-identical mixed vectors in three traces. One full fit: training loss $0.6915 \rightarrow 0.6730$, validation $0.6961 \rightarrow 0.6998$	Whether mixing loses predictive content, or the model fits weak/noisy signals

Root cause: not identified. The news branch is connected; useful generalization remains unproven. A weak effect is not proof of a bug or an unusable dataset. Removing interactions did not rescue performance overall; high vector similarity alone does not justify removing attention.

Next focused step: audit retained versus discarded headline relevance and timing; test any proposed change under a fixed chronological protocol. Possible final-input additions are explicit relevance, novelty and coverage summaries, but their benefit remains untested.

Scope: these development periods were already inspected; subsequent diagnostics/comparisons are exploratory. Paired intervals retain companies together in five-date blocks and omit training uncertainty. Reserved late-2021 outcomes remain unopened. **Sources:** saved CMIN baseline, news-dependency, decay, older-news and new interaction studies; Wang, Cohen & Ma (2024), FININ, Eqs. 1–7; pinned CMIN-US release.